

In T.2579/11 the patent proprietor had not divulged the details of the test protocol applied, citing commercial reasons. The board found that the test results were neither verifiable nor plausible and therefore insufficient to establish that the problem had been solved successfully.

In T.702/99 the board stated that in cases concerning products such as cosmetics, in which applicants or patentees seek to establish that their inventions have an improved "feel" over the prior art, or opponents seek to deny such an improved "feel", it is common for one or more parties to file evidence of comparative tests conducted by a number of persons. It is essential for such tests to be made under conditions which ensure maximum objectivity on the part of those conducting the tests and who may be required at a later date to give evidence in proceedings. It is always desirable that such tests can be shown to be "blind" and that they have been conducted in the strictest conditions; that the testers have had no part in the making of the claimed invention or research leading up to the invention or the patenting procedure. See also T.275/11, T.1962/12, T.165/14 T.2304/16.

In T.172/90 the comparative examples produced did not constitute suitable evidence of inventive step. The board said that the products adduced as a basis of comparison were commercially available and had evidently been selected at random. Technical progress shown in comparison with products of this kind could not be a substitute for the demonstration of inventive step with regard to the closest prior art (referring to T.164/83, OJ 1987, 149). See also T.730/96.

In T.390/88 the board stated that where an alleged invention was prima facie obvious having regard to the prior art, it was however sometimes possible to prove inventiveness by comparative tests showing a significant improvement over the closest prior art. That situation was to be contrasted with other cases, in which it was not prima facie obvious to make the claimed compounds at all, and therefore comparative tests were not essential to establish inventiveness. See also T.656/91, T.60/95, T.930/99.

In T.2319/14 the board disagreed with the respondent's argument that comparative tests always had to be carried out using the closest prior art and that intrinsic comparative tests were not allowed. It was true that, according to the boards' established case law, tests comparing the invention with the prior art had to be conducted in such a way that any effect could be attributed to the distinguishing feature. However, it was also permitted – and might even be necessary – to modify prior-art embodiments in line with the invention to such a degree that the only remaining difference was the feature distinguishing the claim.

In T.1323/17 what counted in the board's opinion was not only whether a causal link between a distinguishing feature over the closest prior art and an effect was demonstrated within the framework of a comparative test submitted by the applicant or patentee. It also counted whether the variant of the closest prior art selected as a reference (or comparative) example for the comparative test was representative of the closest prior art, in the sense that the effect shown to be caused by the distinguishing feature in the context of the comparative test could also be expected to take place within the framework of the closest prior art despite the existence of differences vis-à-vis the reference example of the comparative test.

4.3.3 "Post-published evidence" and the question of whether it was made plausible already by the disclosure in the application that the technical problem is solved

According to the established case law of the boards of appeal the assessment of inventive step is to be made at the effective date of the patent on the basis of the information in the patent together with the common general knowledge then available to the skilled person (T 1329/04, T 1545/08, T 1433/14, T 488/16, T 1322/17).

Post-published evidence (evidence not public before the filing date of the patent in suit and filed after that date; see e.g. T 116/18, OJ 2022, A76) to support that the claimed subject-matter solves the problem to be solved is taken into account if it is already plausible from the disclosure in the application as filed that the problem is indeed solved. In other words, supplementary post-published evidence may not serve as the sole basis to establish that the problem is solved (T 1329/04, T 415/11, T 1791/11, T 488/16, T 212/17, T 1322/17). Post-published evidence can only be used to back up the teaching derivable from the application (see e.g. T 716/08, T 578/06). See also in this chapter I.D.4.4.3.b) "Subsequently invoked technical effect" and also in this chapter I.D.9.9.3 "Broad claims".

In T 1329/04 the board held that the definition of an invention as being a contribution to the art, i.e. as solving a technical problem and not merely putting one forward, required that it was at least made plausible by the disclosure in the application that its teaching indeed solved the problem it purported to solve. In the case in hand since the post-published evidence was considered to be the first disclosure going beyond speculation, it was not taken into consideration (see also T 778/08).

In T 716/08 the board stated that as to the quality of the evidence, "absolute proof" of the achievement of an effect is not required for the effect to be "plausible". See also T 266/10, in which the board similarly held that whether the effect of proposed modifications could be regarded as plausible did not depend on such absolute proof that the effect was actually achieved.

In T 578/06 the board stated that the EPC required no experimental proof for patentability and considered that the disclosure of experimental data or results in the application as filed and/or post-published evidence was not always required to establish that the claimed subject-matter solved the objective technical problem. This was in particular true in the absence of any formulated substantiated doubt. The board re-emphasised in this context however that the case law (with specific reference to T 716/08) considered the establishment of plausibility only relevant when examining inventive step if the case in hand allowed the substantiation of doubts about the suitability of the claimed invention to solve the technical problem addressed and when it was thus far from straightforward that the claimed invention solved the formulated problem. See also T 2197/09.

In T 1322/17 the board noted that as a matter of principle, experimental evidence was not limited to clinical data. It was also noted that experimental evidence was not always necessary to render a certain effect plausible. A mechanistic explanation and/or common general knowledge could be sufficient in certain instances. However, results only known to the inventors derived from studies of unknown set-up (e.g. dosage regimen) could not

be considered when assessing the plausibility of certain effects. A mere statement that a certain effect arises (under conditions that were not reflected by the technical features of the claim), in the absence of any supporting circumstances, did not render the achievement of the effect plausible.

In T. 919/15 the board found that, in the absence of any indications to the contrary in common general knowledge, a synergy between the herbicide combinations containing herbicide (A) which had not been tested in the application as filed could not be regarded as implausible per se. This finding was line with the decision in T. 863/12, in which one of the reasons for considering an effect plausible had been that there was nothing in common general knowledge to cast doubt on its plausibility. It was also not at odds with the decision in T. 1329/04, because it had dealt with a completely different scenario.

In T. 1336/04 the board pointed out that the situation in the case in hand was different from that underlying the decision T. 1329/04 as to the quality of evidence provided in the patent in suit relating to the claimed invention being a bona fide solution to the problem to be solved. In this earlier case, it was not accepted that the then claimed polypeptide SEQ ID No.3 was a member of the TGF-beta family because it had not been shown to have any function, its structure did not conform to that expected from members of the family and the expected sequence homology to previous members of the family was not present. In the case in hand the subject-matter of claim 1 indeed related to novel cellulose- or hemicellulose-degrading enzymes as characterized by the homology of their CBDs to that of known cellulases. On this basis, the board accepted that the problem was satisfactorily solved, also taking into consideration the disclosure in a post-published document.

In T. 1642/07 the board observed that there was no requirement in the EPC, let alone in Art. 56 EPC, that a patent application should include experimental evidence in support of patentability or a claimed technical effect. Hence, the fact that the disclosure in a patent application was merely theoretical and not supported by experimental data was in itself no bar to patentability or to the presence of a technical effect being acknowledged. The board found that the post-published documents could be viewed as being a mere confirmation of the technical effect already announced (albeit at a theoretical level) in the application as filed. It saw no reasons for doubting the described technical effect, and held the post-published evidence could be taken into account.

In T. 536/07 the board noted that although there were no working examples for the claimed subject-matter in the contested patent and it was not disclosed as a preferred embodiment, there was a priori no reason for the skilled person to consider it not to be a plausible solution to the technical problem. There was no indication of a possible prejudice in the art or of foreseen difficulties in carrying out the proposed solution. This situation differed from that underlying decision T. 1329/04, wherein the board decided that the claimed subject-matter did not provide a plausible solution to the identified technical problem. The board in the case in hand took into account the post-published evidence on file demonstrating the feasibility of the proposed solution.

In T. 488/16 the board stated that the question of whether an invention is plausible cannot be answered in general, as this assessment depends upon the individual circumstances,

namely the nature of the invention, the disclosure of the specification and common general knowledge. In the case in hand the board found that the post-published documents were the first disclosure showing that at least for certain thiazole, in particular dasatinib, the purported technical problem had actually been solved. In accordance with established case law, these documents were therefore not taken into consideration in the assessment of inventive step.

In T 2371/13, however, the board held that the fact that an effect had to be regarded as implausible because it was not backed up in the application was not a good enough reason to disregard comparative tests filed later with a view to proving it. To do so for that reason would run counter to the problem and solution approach, which required that a technical problem be defined in light of the closest prior art, which was not necessarily that cited in the application. It was common, the board observed, to try to demonstrate inventive step on the basis of a technical effect not explicitly mentioned in the application as filed. The objection that the invention had yet to be achieved when the application was filed was anyway more a matter for consideration under Art. 83 EPC.

In T 184/16 the board acknowledged plausibility when assessing sufficiency of disclosure and inventive step. It therefore decided to take into account post-published evidence D4. The board noted that acknowledging plausibility was not in contradiction to the finding that the claimed subject-matter was nonobvious in view of the prior art. The criteria for plausibility and obviousness were different. On the one hand, for plausibility of a claimed effect to be acknowledged, it was enough if there were no prima facie serious doubts that the effect could be obtained and, conversely, no a priori reason or indication in the common general knowledge that the effect could not be obtained. On the other hand, obviousness was decided within the framework of the problem-solution approach, where generally an important consideration was whether the claimed solution was suggested and thus made obvious by the prior art.

Some examples of cases, in which the board **took into account** the submitted post-published evidence, are: T 433/05, T 294/07, T 108/09, T 2134/10, T 1677/11, T 872/13, T 1898/15, T 212/17.

In T 108/09 in acknowledging plausibility the board contrasted the case underlying that decision with that in T 1329/04, in which decision there had been prima facie serious doubts that the subject matter solved the problem of the invention. The patent in suit, however, contained detailed information as to how fulvestrant had to be formulated and administered in order to obtain the desired effect as a third-line agent in the treatment of breast cancer. The post-published data of document (10) provided sufficient evidence for demonstrating that the problem had been plausibly solved.

In T 1677/11 the board noted the facts of the present case differed substantially from those underlying decision T 1329/04. In the case in hand the structure of the claimed sodium salt of (-)-omeprazole was fully consistent with that of the known class of gastric acid secretion inhibitors, whereas in T 1329/04 the structural features of the polypeptide were found to be inconsistent with that expected of the superfamily. Moreover, the patent in suit disclosed a synthesis of the claimed salt and provided a clear statement that it provided

"an improved therapeutic profile such as a lower degree of interindividual variation". The board thus saw no reason a priori for the skilled person to regard it as being implausible and considered it appropriate to take into account the post-published evidence submitted for the purpose of assessing whether or not the effect identified was indeed observed.

Some examples of cases, in which the board **did not consider** the submitted post-published evidence, are: T 1306/04, T 861/08, T 1791/11, T 125/12, T 1196/12, T 1433/14, T 488/16, T 1322/17.

In T 1791/11 the board stated it was apparent from the patent application itself that it was not yet known which variants solved the problem and that a test still had to be performed to confirm the alleged advantage. The board thus came to the conclusion that the patent did not render it plausible that the claimed subject-matter solved the technical problem as formulated by the appellant-proprietor, and the experimental post-published evidence of was in fact the sole basis allowing to conclude that said problem has been plausibly solved.

In T 116/18 (OJ 2022, A76) the board noted that whether post-published evidence could be taken into account was a fundamental question of law for which diverging lines of case law existed. There were, in the board's view, **three diverging lines of case law**, two of which contained extreme positions: one being a strict application of the ab initio plausibility standard (point 13.4 of the Reasons) and the other one applying the no plausibility standard (point 13.6 of the Reasons). The ab initio implausibility standard in terms of its results appeared to the board to lie somewhere between these two extreme lines of case law (point 13.5 of the Reasons). The board referred the following questions to the Enlarged Board of Appeal (to be decided in **G 2/21**):

If for acknowledgement of inventive step the patent proprietor relies on a technical effect and has submitted evidence, such as experimental data, to prove such an effect, this evidence not having been public before the filing date of the patent in suit and having been filed after that date (post-published evidence):

1. Should an exception to the principle of free evaluation of evidence be accepted in that post-published evidence must be disregarded on the ground that the proof of the effect rests exclusively on the post-published evidence?
2. If the answer is yes, can the post-published evidence be taken into consideration if, based on the information in the patent application in suit or the common general knowledge, the skilled person at the filing date of the patent application in suit would have considered the effect plausible (ab initio plausibility)?
3. If the answer to the first question is yes, can the post-published evidence be taken into consideration if, based on the information in the patent application in suit or the common general knowledge, the skilled person at the filing date of the patent application in suit would have seen no reason to consider the effect implausible (ab initio implausibility)?

Chapter II.C.6.8 "Post-published documents" deals with the topic of post-published evidence with regard to sufficiency of disclosure (Art. 83 EPC).